

# Predicting 30-day Diabetic Hospital Readmission: A Replication Study

Nellia Kornilova  
University of Nottingham

**Abstract**—This replication study builds on Emi-Johnson and Nkrumah (2025) [1] to predict 30-day diabetic hospital readmission. Using the UCI Diabetes dataset (N = 101,766 after cleaning), XGBoost achieved a superior 0.680 AUC-ROC and 0.897 recall. SHAP analysis identified prior inpatient visits and medication burden as primary predictors. High recall but low accuracy and precision in my replication, and lower recall but higher accuracy and precision in the original study, highlight a critical precision-recall trade-off that remains a barrier to clinical adoption of such models.

## I. INTRODUCTION AND EDA

### A. Problem and dataset explained

Diabetes remains one of the most significant global health challenges, characterized by its chronic nature and the high complexity of clinical management. Within the inpatient setting, diabetic patients often face constant readmissions. Consequently, 30-day hospital readmission has emerged as a benchmark for healthcare quality and patient safety. The adoption of Electronic Health Records (EHR) has provided a repository of structured data, offering an opportunity to move toward automated, evidence-based predictive modelling to help hospitals forecast which patients might return and hence adequately plan resources.

This study is a replication and extension of the research conducted by Emi-Johnson and Nkrumah [1]. It seeks to address the following core research question: **‘To what extent can electronic health record (EHR) data, specifically demographics, admission details, and medication regimens, be used to predict early (<30-day) hospital readmission for diabetic patients using supervised machine learning algorithms?’**. By replicating the methodology of Emi-Johnson and Nkrumah, this project compares the efficacy of baseline models against ensemble methods and neural networks (DNNs) to determine the most effective architecture for handling imbalanced clinical outcomes.

The research utilizes the UCI Diabetes 130-US Hospitals dataset, which consists of de-identified longitudinal records from 130 United States hospitals [2]. Collected between 1999 and 2008, the dataset contains 101,766 unique encounters for patients diagnosed with diabetes. Each record represents an inpatient stay lasting between one and 14 days where diabetes was recorded as a diagnosis. The data includes more than 50 clinical and demographic features, such as race, gender, age, number of lab procedures, and time spent in the hospital. It also tracks historical medication use, specifically changes in dosage for 23 distinct. The target variable for prediction is the readmission status, specifically isolating cases where the patient returned to the hospital within a 30-day window.

### B. Exploratory Data Analysis

The Exploratory Data Analysis (EDA) focused on identifying the underlying distributions of clinical features and the strength of associations between patient history and 30-day readmission.

*a) Univariate Analysis and Distributions* (Figure 1): The distribution overlays for key clinical and aggregated features show a cohort heavily weighted toward older populations and varying degrees of care intensity. The age distribution, characterized by spikes due to the conversion of age ranges to midpoints, peaks in the 70–80-year old category, aligning with the expected demographic for diabetic inpatient encounters. Clinical intensity metrics, such as ‘time\_in\_hospital’ and ‘num\_medications’ show a right skew. While most patients remained hospitalized for fewer than five days,

a significant tail exists for longer stays, representing more complex cases. A large volume of the records shows zero prior encounters, which contributes to the class imbalance observed in the ‘readmitted’ target variable.

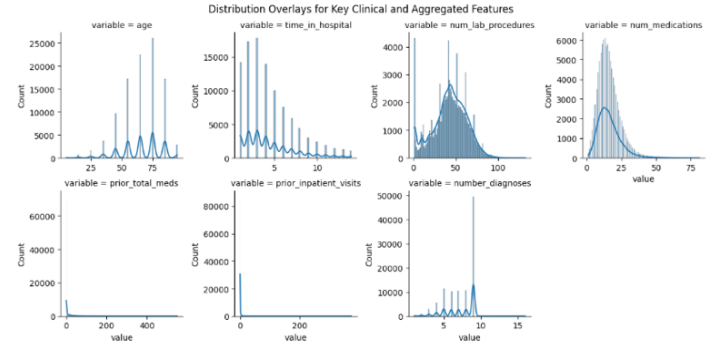


Figure 1. Univariate analysis and distributions

*b) Multivariate Analysis and Correlation* (Figure 2): The correlation heatmap provides a statistical justification for the feature engineering approach and model selection. A primary insight is the high degree of multicollinearity among the aggregated patient history features. Strong positive correlations exist between ‘prior\_total\_meds’ and ‘prior\_total\_labs’ ( $r=0.99$ ) as well as between ‘prior\_days\_in\_hospital’ and ‘prior\_total\_meds’ ( $r=0.99$ ). These values indicate that laboratory testing and medication prescriptions are synchronized over a patient's historical encounter sequence. This multicollinearity supports the decision to use ensemble frameworks like XGBoost and Random Forest, which are more tolerant to redundant features.

Furthermore, the heatmap demonstrates that ‘prior’ features have a stronger association with the ‘readmitted’ target than single-encounter variables. ‘Prior\_inpatient\_visits’ ( $r=0.14$ ), ‘prior\_total\_meds’ ( $r=0.13$ ), and ‘prior\_days\_in\_hospital’ ( $r=0.13$ ) all show higher Spearman correlation coefficients with readmission than ‘time\_in\_hospital’ ( $r=0.05$ ) or ‘num\_medications’ ( $r=0.05$ ). This confirms the clinical logic that a patient's historical frequency of hospitalization and chronic medication burden are more significant predictors of near-term readmission than the specifics of a single isolated stay. Consequently, the EDA validates the inclusion of aggregated rolling histories to improve the discriminative ability of the predictive models.

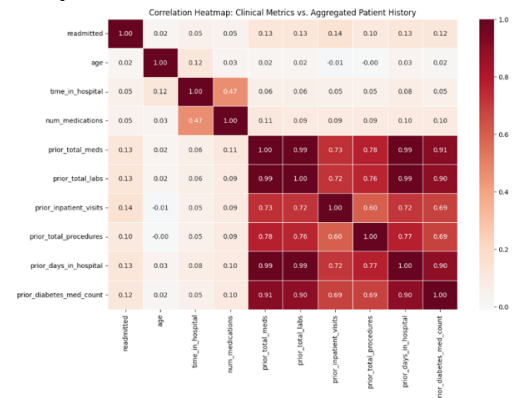


Figure 2. Multivariate analysis and correlation

## II. RELATED WORK

Predicting 30-day hospital readmission for diabetic patients has become an important point in machine learning due to its status as primary indicator of healthcare quality and institutional efficiency (in the US). As discussed in the previous section, this study replicates and build upon the research of Emi-Johnson and Nkrumah (2025), which evaluated four ML models - Logistic Regression, Random Forest, XGBoost, and Deep Neural Networks (DNNs) - using the UCI Diabetes 130-US Hospitals dataset. Their work demonstrated that ensemble models, specifically XGBoost outperform traditional baselines by achieving a higher Area Under the Receiver Operating Characteristic curve (AUC-ROC) of 0.667.

The wider literature confirms the effectiveness of gradient-boosting frameworks like XGBoost for structured clinical datasets which is cited as superior model in the original study. Proposed by Chen and Guestrin (2016) [3], XGBoost is indeed recognized for its ability to handle sparse data and model complex interactions, which is essential for EHR attributes like diagnosis codes and medication changes. Furthermore, since interpretability is a prerequisite for clinical adoption, the original study utilizes SHapley Additive exPlanations (SHAP). While the underlying principles of coalitional game theory were established by Lloyd Shapley (1953) [4], the SHAP framework was introduced by Lundberg and Lee (2017) [5] to bring these concepts into machine learning. This approach allows for detailed interpretability by quantifying each clinical feature's contribution to an individual patient's risk score, which is central to the transparency of clinical machine learning studies.

## III. METHODOLOGY

The approach to methodology was mostly following the paper I was replicating with some necessary changes.

### A. Data Cleaning

The dataset contained '?' to indicate missing data. All the values with '?' were replaced with NaN. The features where most of the values were missing, such as 'weight' (97% missing) and 'payer\_code' (40 % missing) were dropped. Individual drugs were also dropped since they provided little value for the predictive efficiency as their total number was contained in another 'num\_medications' column.

### B. Feature Engineering and Aggregation

Following clinical logic, ICD-9 diagnosis codes were grouped into categories (e.g. Circulatory, Respiratory, Diabetes, etc.) and less numerous clinical specialties were grouped into 'Other'. The new features were also created using aggregation by patient (cum\_sum) to reflect the accumulation of care intensity across multiple visits. This decision was taken to reflect care intensity over multiples visits. This specific methodology was chosen to avoid temporal data leakage as it is using a rolling history rather than a global sum that would introduce leakage. The 'age' column which contained age ranges column was formatted to capture only the middle point of the range.

After the data was divided to 80% train and 20% test, 'ColumnTransformer' was implemented to handle heterogeneous data. Categorical features with missing values were firstly processed with 'SimpleImputer' - mode imputation because this is the most appropriate method for dealing with categorical missing values since neither mean nor median can be calculated on such values. These processed categorical values were then encoded with 'OneHotEncoder'. Other categorical values were sent through one-hot-encoding process separately. Numerical values were scaled with 'StandardScaler'. By using a ColumnTransformer inside a pipeline (pipeline is explained in the next subsection), I ensure that

parameters for preprocessing, such as the mean and standard deviation for StandardScaler or the most frequent value for mode imputation, are calculated only on the 80% training data to prevent data leakage.

It is worth noting that in the paper I am replicating, the missing values were dealt with using Mean Imputation as they stated that the missing values were numerical [1]. In my analysis I did not encounter numerical missing values outside of the columns that were dropped and hence this method was not applied. In their paper they don't mention how they processed categorical missing values.

### C. Model Development and Evaluation

Following the paper, I implemented four models: logistic regression, random forest, XGBoost and DNNs. These models were chosen by authors for the following reasons: logistic regression is a good baseline due to the ease of interpretability, random forest is 'robust to nonlinear interactions', XGBoost is an advanced gradient-boosting framework used for its efficiency as an ensemble learning method and DNNs is designed for complex non-linear relationships data [1, 3]. Each model was implemented using 'Pipeline' method in scikit-learn. Addressing class imbalance in the test dataset due to smaller amount of readmission cases (Figure 3), the models utilized class weighting, distributing higher weights to less frequent class instances, mirroring the original paper.

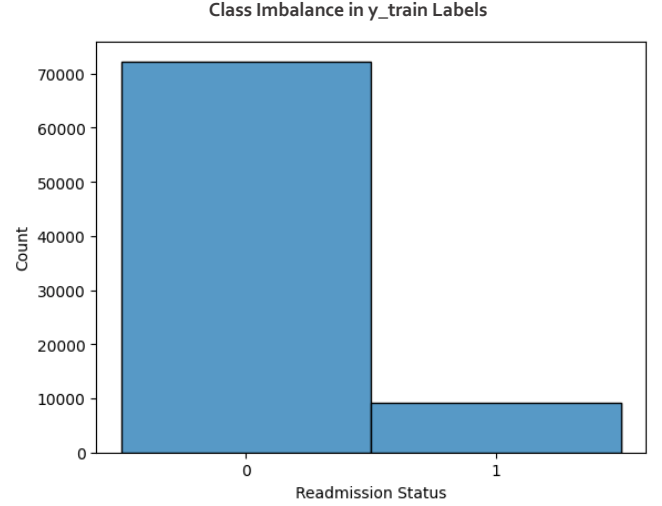


Figure 3. Class imbalance in labels (y\_train): 1 - 'readmitted within 30 days'; 0 - 'not readmitted'

Each model was initially validated using common metrics like accuracy, precision, recall, F1 score and AUC-ROC. Confusion matrix were also plotted individually for each model (Figure 4). AUC-ROC and Precision-Recall curves were plotted for all models on a single plane for the ease of comparison (Figure 5-6). To replicate the methods of the paper as closely as possible SHAP analysis was also used on XGBoost to evaluate the most important features through interpretability (Figure 7).

### D. Hyperparameter Tuning

To optimize the performance, I used 'GridSearchCV' and 'RepeatedStratifiedKFold' from the scikit-learn library. This approach employed a three-fold three-repeats stratified cross-validation strategy on the training dataset to ensure that each parameter combination was evaluated across different data folds while maintaining the original class distribution. The best parameters were chosen based on the AUC-ROC score as this is a more appropriate metric with imbalanced classes compared with accuracy since accuracy can be artificially inflated by a model that simply predicts the majority class for every patient [6].

### E. Statistical Testing

Wilcoxon signed-rank test is used to compare the performance of the baseline logistic regression and XGBoost since the performance of these two models is the most significantly different. This test is chosen because it is ‘distribution-free’ meaning it does not look at raw values but rather checks the ranks of the differences and this fits well with limited folds in my cross-validation process (due to limited available compute).

## IV. RESULTS

### A. Overview

As previously mentioned, the final dataset contained 101,766 records. The class distribution in  $y_{train}$  was significantly imbalanced, with 72273 non-readmissions and 9139 readmissions within 30 days (Figure 3). This resulted in a readmission rate of 11.2%, consistent with the original study.

### B. Model Performance Metrics

Logistic regression served as the baseline model, achieving an accuracy of 0.674 and an AUC-ROC of 0.642. It has a recall of 0.511 with low precision of 0.169 (Figure 4). This suggests that the model identified half of the readmission cases correctly, but also produced a high volume of false positives.

Random Forest and XGBoost achieved near-perfect recall, 0.911 and 0.897, respectively, but at the expense of extremely low accuracy, 0.283 and 0.338. This behavior indicates that the balanced class weighting optimization may be overoptimized to predict the minority class (readmission) for nearly every instance in the test set. Despite the low accuracy, XGBoost achieved the highest AUC-ROC of 0.679, slightly outperforming the 0.667 AUC-ROC reported in the original paper. The Wilcoxon test used in my study confirmed that the XGBoost model outperforms Logistic Regression baseline across 10 folds. This improvement was statistically significant as p-value is less than 0.05 ( $p=0.019$ ).

DNNs showed the highest raw accuracy at 0.854 and the highest precision at 0.240. However, it suffered from a low recall of 0.153, indicating a tendency to predict the majority class (non-readmission). Its AUC-ROC of 0.650 was higher than the 0.579 reported for the DNNs in the original study.

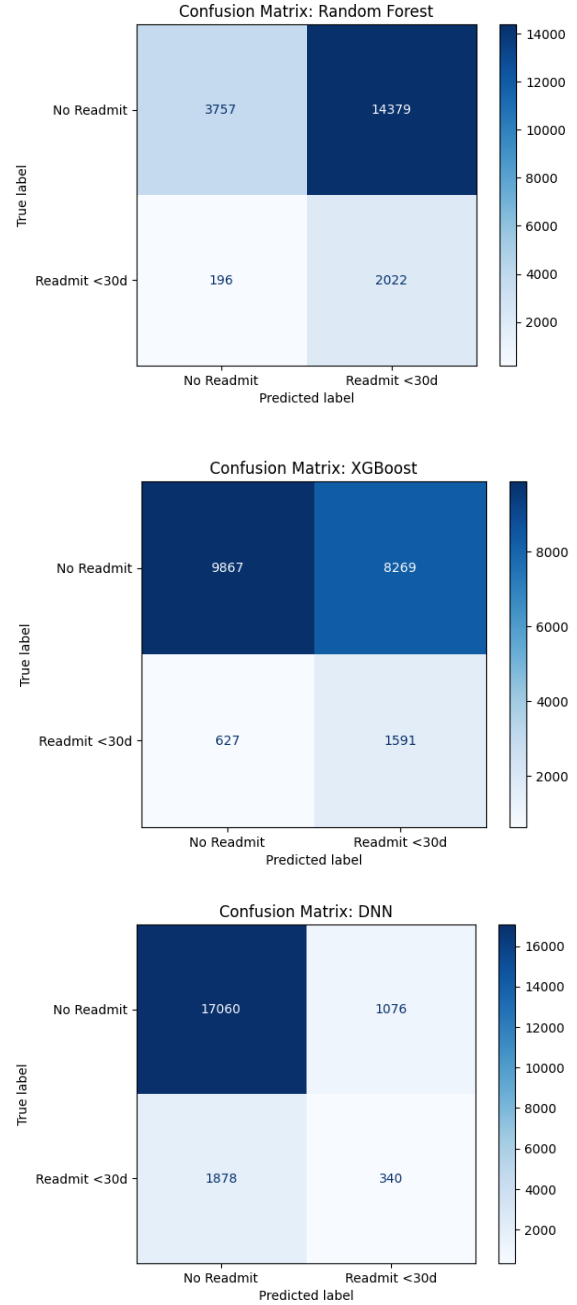
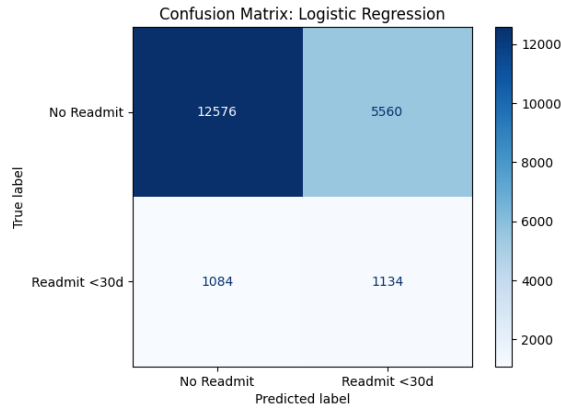


Figure 4. Confusion Matrix for all models



### C. Comparison with the Original Study

The replication results show consistent AUC-ROC rankings but divergent accuracy, precision and recall metrics compared to Emi-Johnson et al. The original study reported high accuracy (~0.89) across logistic regression, random forest, and XGBoost, but extremely low recall for the readmission class (0.002 to 0.017). In contrast, the current replication's ensemble models (random forest and XGBoost) prioritised recall, achieving values near 1.0, which likely led to a drop in accuracy and precision. The XGBoost model remained the strongest model in both studies, consistently achieving the highest AUC-ROC (Figure 5).

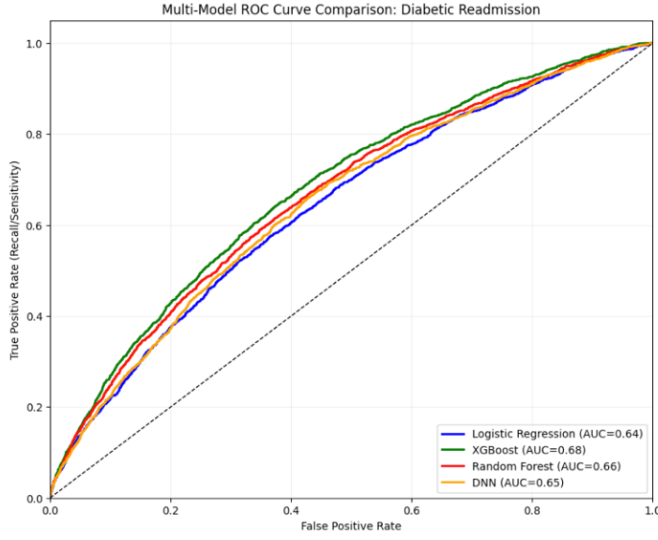


Figure 5. AUC-ROC curves comparison

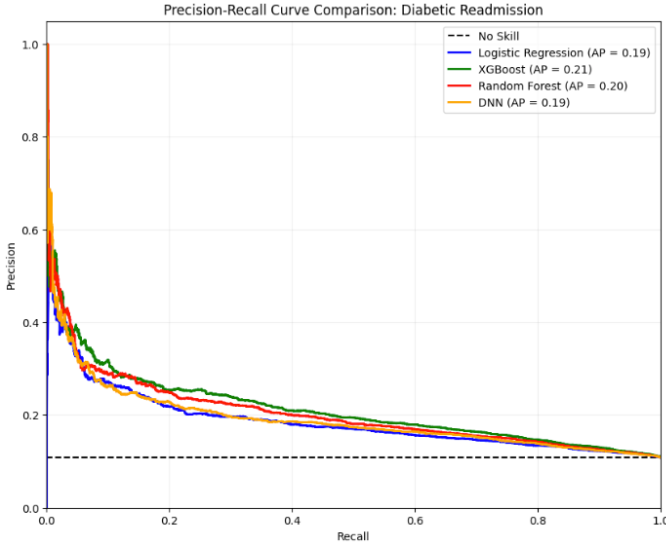


Figure 6. Precision-Recall curve for all models

## V. DISCUSSION

### A. Clinical Interpretation

The results of this replication demonstrate that the XGBoost architecture is superior for predicting 30-day readmissions. This aligns with Emi-Johnson et al. study confirming the model's suitability for high-dimensional tabular EHR data. The primary divergence between the studies lies in the precision-recall trade-off (Figure 6). My implementation utilised more aggressive class weighting thus prioritising the identification of at-risk patients (recall), leading to a significant drop in accuracy for ensemble methods. Clinically, this represents a shift from a safer model that rarely predicts readmission to a sharper model that predicts almost all 'at-risk' patients but with a trade-off of also misclassifying the ones that are not at risk.

My results reinforce the current consensus in machine learning literature that gradient-boosting frameworks like XGBoost often outperform deep learning for structured EHR datasets [7-9]. The AUC-ROC values obtained (0.64-0.68) are consistent with the difficulty of readmission prediction, which rarely exceeds 0.70 in similar studies due to noise in and missing social health determinants in standardized EHR datasets, as noted by Goldstein et al. [8]. The inclusion of aggregated patient history features in my study appears to have strengthened DNN's discriminative ability compared to the original study, as evidenced by the increase in

AUC-ROC. This supports the research by Rajkomar et al. referenced in the original paper, which emphasizes the value of longitudinal patient data over a single encounter snapshot [9]. SHAP analysis was also consistent with the original paper showing that discharge destinations, previous admissions and medication burden were the most influential predictors of readmission (Figure 7).

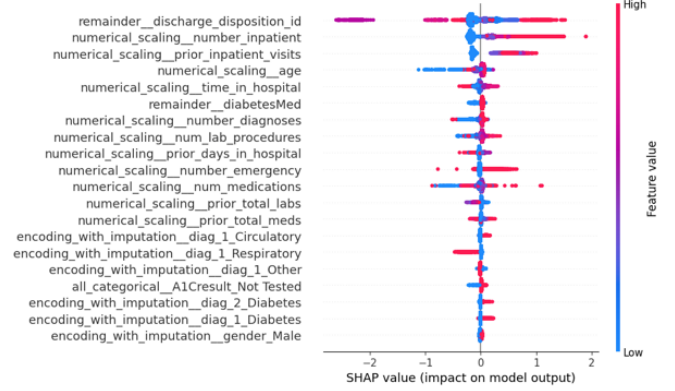


Figure 7. SHAP analysis

### B. Limitations

There are several limitations in this replication which extend beyond those discussed in original study. First, the extreme recall values for Random Forest and XGBoost make them practically unstable in the real-world hospital scenario, despite the satisfactory AUC-ROC performance, as the false positive rate would overwhelm clinical resources. Second, the computational constraints mentioned in the methodology section necessitated a limited cross-validation strategy (three-folds, three-repeats), which may affect the generalizability of the optimal hyperparameters.

Moreover, the feature engineering method I employed, specifically the use of cumulative sums for care intensity, relies on the chronological integrity of the dataset. Hence, if there are any errors in the recording of encounter sequences, a temporal bias might be introduced. Also, the cleaning strategy I used, which is more extensive than in the original study may have simplified the underlying data structure, potentially leading to inflated performance metrics for some models relative to the original noisier dataset.

## VI. CONCLUSION

This research confirms that gradient-boosting frameworks, specifically XGBoost, provide superior performance for readmission prediction using structured, high-dimensional EHR data. The study successfully replicated the model rankings of previous work with minor differences discussed in the previous section.

A key finding of this extension is that feature engineering focused on rolling care intensity and historical encounter frequency provides higher predictive value than single-encounter snapshots. This was particularly evident in the 0.650 AUC-ROC achieved by the DNNs, which exceeded previous benchmarks. However, the precision-recall trade-off remains a problem as the high false-positive rates observed in the high-recall ensemble models could potentially overwhelm clinical resources in a real-world setting. The AUC-ROC scores range (0.64-0.68) across all models highlight the natural noise in EHR datasets and the likely influence of unobserved social determinants of health.

Future iterations should investigate advanced sampling techniques like SMOTE and the integration of external socioeconomic data to enhance model precision and facilitate clinical adoption.

## REFERENCES

- [1] O. G. Emi-Johnson and K. J. Nkrumah, "Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic health record data," *Cureus*, vol. 17, no. 4, p. e82437, Apr. 2025. doi: 10.7759/cureus.82437.
- [2] B. Strack et al., "Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records," *BioMed Res. Int.*, vol. 2014, Art. no. 781670, 2014. doi: 10.1155/2014/781670.
- [3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794. [Online]. Available: <https://doi.org/10.1145/2939672.2939785>
- [4] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games (AM-28)*, vol. II, H. W. Kuhn and A. W. Tucker, Eds. Princeton, NJ: Princeton Univ. Press, 1953, pp. 307–317.
- [5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 4768–4777.
- [6] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognit. Lett.*, vol. 27, no. 8, pp. 861–874, Jun. 2006. doi: 10.1016/j.patrec.2005.10.010.
- [7] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794. [Online]. Available: <https://doi.org/10.1145/2939672.2939785>
- [8] B. A. Goldstein et al., "Opportunities and challenges in developing risk prediction models with electronic health records data: A systematic review," *J. Amer. Med. Inform. Assoc.*, vol. 24, no. 1, pp. 198–208, Jan. 2017. doi: 10.1093/jamia/ocw042.
- [9] A. Rajkomar et al., "Scalable and accurate deep learning with electronic health records," *npj Digit. Med.*, vol. 1, no. 1, p. 18, May 2018. doi: 10.1038/s41746-018-0029-1.D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, arXiv:1312.6114. [Online]. Available: <https://arxiv.org/abs/1312.6114>. 1038/s41746-018-0029-1.D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, arXiv:1312.6114. [Online]. Available: <https://arxiv.org/abs/1312.6114>